

Theory Questions :-

- Searching Algorithms Interview Questions
Linear Search

Q1. What is linear search?

Q2. What is the time complexity of linear search?

Q3. When is linear search preferred?

Q4. What are the advantages and disadvantages of linear search?

Binary Search

Q5. What is binary search?

Q6. What are the prerequisites for binary search?

Q7. What is the time complexity of binary search?

Q8. Difference between iterative and recursive binary search

Q9. Why is binary search faster than linear search?

Q10. What are common mistakes in binary search implementation?

* Sorting Algorithms Interview Questions

Q1. What is sorting? Why is it important?

Q2. Types of sorting algorithms

Q3. Difference between stable and unstable sorting

Difference between in-place and out-of-place sorting

Algorithm-Specific Questions

♦ Bubble Sort

How does bubble sort work?

Time complexity of bubble sort

♦ Selection Sort

How does selection sort work?

Difference between selection sort and bubble sort

♦ Insertion Sort

How does insertion sort work?

When is insertion sort preferred?

♦ Merge Sort

How does merge sort work?

Why is merge sort called divide and conquer?

♦ Quick Sort

How does quick sort work?

What is pivot element?

Worst case of quick sort

♦ 6. Advanced / Conceptual Questions

Which sorting algorithm is best and why?

Can you sort without extra space?

What is divide and conquer approach?

Difference between comparison-based and non-comparison sorting
When would you use which sorting algorithm?

MCQ Questions :-

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Q1. What will be the output of the following Java code?

```
int arr[] = {5, 2, 8, 1};
for(int i=0; i<arr.length-1; i++)
{
    for(int j=0; j<arr.length-i-1; j++)
    {
        if(arr[j] > arr[j+1])
        {
            int temp = arr[j];
            arr[j] = arr[j+1];
            arr[j+1] = temp;
        }
    }
}
System.out.print(arr[2]);
```

- A) 1
- B) 2
- C) 5
- D) 8

Q2. What is the output?

```
int arr[] = {10, 20, 30, 40, 50};
int key = 30;
int low = 0, high = arr.length - 1;
```

```
while(low <= high)
{
    int mid = (low + high) / 2;

    if(arr[mid] == key)
    {
        System.out.print(mid);
        break;
    }
    else if(arr[mid] < key)
        low = mid + 1;
    else
        high = mid - 1;
}
```

- A) 1
- B) 2
- C) 3
- D) 4

Q3. What will be the output?

```
int arr[] = {9, 7, 5, 3};
int key = 5;
int i;
```

```
for(i=0; i<arr.length; i++)
{
    if(arr[i] == key)
        break;
}
System.out.print(i);
```

- A) 1
- B) 2
- C) 3
- D) 4

Q4. What is the output?

```
int arr[] = {4, 3, 2, 1};
```

```
for(int i=0; i<arr.length; i++)
{
    int min = i;

    for(int j=i+1; j<arr.length; j++)
    {
        if(arr[j] < arr[min])
            min = j;
    }

    int temp = arr[i];
    arr[i] = arr[min];
    arr[min] = temp;
}
```

```
System.out.print(arr[0] + " " + arr[3]);
```

- A) 1 4
- B) 4 1
- C) 2 3
- D) 3 2

Q5. What will be the output?

```
int arr[] = {1,2,3,4,5,6,7};
int key = 8;
int low=0, high=arr.length-1;
int flag=0;
```

```
while(low<=high)
{
    int mid=(low+high)/2;

    if(arr[mid]==key)
    {
        flag=1;
```

```

        break;
    }
    else if(arr[mid]<key)
        low=mid+1;
    else
        high=mid-1;
}

```

System.out.print(flag);

- A) 1
- B) 0
- C) 7
- D) Error

Practice Questions :-

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Q1. Write a Java program to perform Linear Search and find an element in an array.

Input:

Array = {10, 25, 40, 55, 70}

Search = 55

Output:

Element found at index 3

Q2. Write a Java program to sort an array using Bubble Sort.

Input:

Array = {8, 3, 6, 1, 9}

Output:

1 3 6 8 9

Q3. Write a Java program to perform Binary Search on a sorted array.

Input:

Array = {5, 10, 15, 20, 25, 30}

Search = 20

Output:

Element found at index 3